GitHub Repository File Key:
June 16, 2026

"R Script"

- **identification_counts_QC.Rmd** - Bar/summary plots of unique protein, peptide, and PSM counts per condition; QC overview of how identification depth changes across DNA Shield dilutions for both Cp34H and T. pseudonana.
- **NSAF_replicate_reproducibility.Rmd** - Pairwise replicate correlation plots ($\log_2$ PSM, Pearson r) and NSAF replicate scatter plots (log-log, density-colored) for both organisms; confirms technical reproducibility across triplicates.
- **Venn_Diagrams_Missingness.Rmd** - Prepares protein detection lists per condition for Venn diagram upload to Venny 2.1.0; visualizes which proteins are shared or unique across DNA Shield dilution groups for both organisms.
- **figure_unique_peptides_nature.Rmd** - Nature-style ridge/beeswarm figure of condition-unique peptide landscapes; compares Cp34H B-CTL vs B-1.5 and T. pseudonana P-CTL vs. P-2.0.
- **venn_unique_peptide_GO_analysis.Rmd** - Venn overlap of unique peptides between control and treatment, with GO enrichment analysis for condition-specific peptide sets (both organisms).
- **metagomics2_GO_analysis.Rmd** - Reads metaGOmics2 v0.1.2 output (go_terms.csv, coverage.csv, go_taxonomy_combo.csv) for bacteria (list_000, list_001) and phytoplankton (list_004, list_005); produces GO term distributions, taxonomy summaries, and NSAF-weighted GO plots.
- **volcano_plots.Rmd** - GO-annotated volcano plots for B-CTL vs B-1.5 and P-CTL vs P-2.0; Welch's t-test on $\log_{10}$ NSAF + BH FDR, points colored by GO biological process category from metaGOmics2 annotations.
- **CP34H_peptide_filtering.Rmd** - Filters raw Limelight/Comet peptide lists for both organisms: removes contaminants (cRAP), applies PSM/peptide $q \leq 0.01$, and writes the cleaned bacteria_peptides_filtered.csv and phytoplankton_peptides_filtered.csv used by downstream scripts.
- **CP34H_DNA_Shield_Analysis.Rmd** - Main analysis notebook for Cp34H across all four conditions (BC, E, F, G); NSAF distributions, protein/peptide counts, dose-response trends.
- **CP34H_limma_analysis.Rmd** - Differential abundance via limma moderated t-test on $\log_2$ NSAF with median centering and BH FDR; outputs Bacteria_4group_limma_results.xlsx.
- **CP34H_supplemental_BC_vs_E.Rmd** - Three supplemental comparisons of BC vs E: NSAF density scatter (log-log), NSAF test of proportions, and PSM test of proportions; run on both the full-elution and 3–24 min elution files.
- **TPs_DNA_Shield_Analysis.Rmd** - Main analysis notebook for T. pseudonana across three conditions (P-CTL, P-1.5, P-2.0); NSAF distributions, identification counts, and dose-response trends mirroring the bacteria notebook.

"Data"

Root-level files

- **2026_Feb_02_ncbi_colwellia_tpseudonana_contam.fasta** - The combined protein sequence database used for Comet database searching: Cp34H + T. pseudonana sequences from NCBI plus the cRAP contaminant database. The date in the filename marks when it was built.
- **bacteria_peptides_filtered.csv** - Post-filtered peptide table for Cp34H. Contaminants and low-confidence IDs removed by CP34H_peptide_filtering.Rmd; this is the input for pairwise correlation plots and replicate QC.
- **phytoplankton_peptides_filtered.csv** - Same as above for T. pseudonana.
- **bacteria_modificaiton_table.txt** - Magnum open-modification search results for Cp34H; rows are modification masses, columns are PSM counts per replicate (B-CTL, 1.5, 1.75, 2.0). Used for Figure 16.
- **phytoplankton_modification_table.txt** - Same as above for T. pseudonana (P-CTL_1-3, P-1.5_1-3, P-2.0_1-3 replicates).

Bacteria/

- **Bactera_Comet_all_samples_proteins.txt** - Full-elution Limelight/Comet protein-level output for Cp34H across all 12 samples (B-CTL_1-3, B-1.5_1-3, B-1.75_1-3, B-2.0_1-3). Contains PSM counts, NSAF values, and protein descriptions for all conditions. Primary input for NSAF analyses and volcano plots.
- **Bacteria_Comet_elution_3_to_24min_prot.txt** - Same as above but restricted to the 3-24 min LC elution window. Used alongside the full-elution file in the B-CTL vs B-1.5 proportions tests.
- **Bacteria_Comet_all_samples_peptides.txt** - Raw peptide-level Limelight/Comet output for Cp34H; input to CP34H_peptide_filtering.Rmd.
- **proteins_clean.csv** - Contaminant-filtered version of the bacteria protein table; used by NSAF_replicate_reproducibility.Rmd for the NSAF replicate scatter plots.

Phytoplankton/

- **Phytoplankton_Comet_all_samples_proteins.txt** - Full-elution protein-level output for T. pseudonana across 9 samples (P_CTL1–3, P-1.5_1.3, P-2.0_4-6). Primary input for phytoplankton NSAF analyses and volcano plots.
- **Phytoplankton_Comet_all_samples_peptides.txt** - Raw peptide-level output for T. pseudonana.
- **proteins_clean.csv** - Contaminant-filtered protein table for T. pseudonana; used for NSAF replicate scatter plots.

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limma_results/

- **Bacteria_4group_limma_results.xlsx** - Output of CP34H_limma_analysis.Rmd: per-protein moderated t-test results (log_2FC, p-value, BH FDR) for all pairwise comparisons among the four Cp34H conditions.

- **Phytoplankton_limma_results.xlsx** - Same for T. pseudonana across its three conditions.

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venn_upload_files/

Seven CSVs of protein accession lists, one per condition, prepared by Venn_Diagrams_Missingness.Rmd for upload to the Venny 2.1.0 web tool (Venn diagrams are not generated in R):

- **Bacteria_BC.csv** - Cp34H control detected proteins (~18,400 entries)
- **Bacteria_E.csv** - Cp34H B-1.5
- **Bacteria_F.csv** - Cp34H B-1.75
- **Bacteria_G.csv** - Cp34H B-2.0
- **Phytoplankton_C.csv** - T. pseudonana control
- **Phytoplankton_D1.5.csv** - T. pseudonana P-1.5
- **Phytoplankton_D1to2.csv** - T. pseudonana P-2.0